Arash Golmohammadi

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Summary

Computational engineer and PhD candidate specializing in applied mathematics and numerical methods, stochastic modeling, and machine learning for biophysical systems. Experienced in data-driven models, simulation design, and automated pipelines in hydrodynamics, finance, and medical sciences.
Key competences: Mathematical modeling, Physics, Deep learning, Optimization, Complex systems

Work Experience

2020.12 - 2022.06 **Research Assistant** SFB 1270 ELAINE, University of Rostock
Implemented an exact numerical solver for predicting the electric potential in deep brain stimulation; Application in neurodegenerative motor diseases such as Parkinson's disease.

2020.02 - 2020.07 **Research Intern** MCSS lab, EPFL
Curated a database of numerically simulated turbulent flow around 2D turbine blades; Application in training deep network-based controllers for monitoring drones.

2018.07 - 2019.02 **Machine Learning Intern** Deeption SA
Optimized and automated algorithmic equity trading in New York Stock Exchange market. Enabled live trading by communicating with broker, and providing technical support for the trading hardware.

Education

2022 - present	Ph.D. in Computational Neuroscience at **University of Göttingen**	
2017 - 2020	M.Sc. in Computational Science and Engineering at **EPFL**	(GPA: 5.0/6.0)
2012 - 2017	B.Sc. in Physics & Mechanical Engineering (double major) at **Sharif University of Technology**	(GPA: 18.05/20)

Awards

2012 **Gold medal** at the International Olympiad on Astronomy and Astrophysics (IOAA)

Skills

Technical	Programming (Python, C, C++, MATLAB, HTML, CSS), git, Unix, Latex, MS Office
Scientific	Deep learning, Numerical analysis, Statistical physics, Digital signal processing, Statistics
Languages	English (C1), German (B1), French (A1), Persian (Native)

Other Activities

Community work	Treasurer at the Iranian Association at EPFL (2019), contributing to open-source projects
Hobbies	Classical and film music, podcasts, reading novels, watching movies, cycling, playing piano, automation and integration, gadgets building with microprocessors
Passions	Digital democracy platforms, Computational social sciences

References

Available upon request.